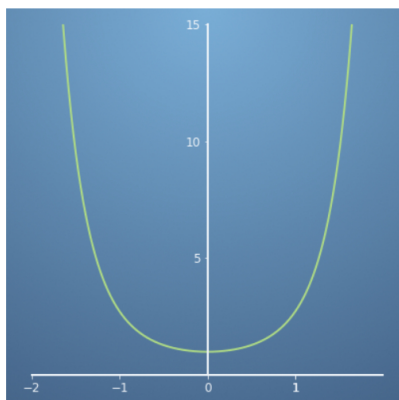


1. In the two previous videos, we have shown the short mathematical proofs for the Taylor series, and for special cases when the starting point is $x = 0$, the Maclaurin series. In these set of questions, we will begin to work on applying the Taylor and Maclaurin series formula to obtain approximations of functions.

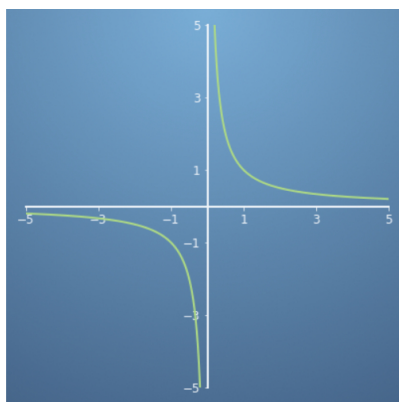
1 point



For the function $f(x) = e^{x^2}$ about $x = 0$, using the the Maclaurin series formula, obtain an approximation up to the first three non zero terms.

- ☐ $f(x) = x^2 + \frac{x^4}{2} + \frac{x^6}{6} + \dots$
- ☒ $f(x) = 1 + x^2 + \frac{x^4}{2} + \dots$
- ☐ $f(x) = 1 - x^2 - \frac{x^4}{2} \dots$
- ☐ $f(x) = 1 + 2x + \frac{x^2}{2} + \dots$

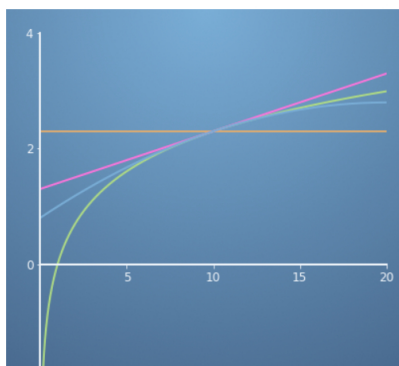
1 point



2. Use the Taylor series formula to approximate the first three terms of the function $f(x) = 1/x$, expanded around the point $p = 4$.

- ☐ $f(x) = -\frac{1}{4} - \frac{(x+4)}{16} - \frac{(x+4)^2}{64} \dots$
- ☐ $f(x) = \frac{(x-4)}{16} + \frac{(x-4)^2}{64} - \frac{(x-4)^3}{256} \dots$
- ☒ $f(x) = \frac{1}{4} - \frac{(x-4)}{16} + \frac{(x-4)^2}{64} + \dots$
- ☐ $f(x) = \frac{1}{4} - \frac{(x+4)}{16} + \frac{(x+4)^2}{64} + \dots$

1 point

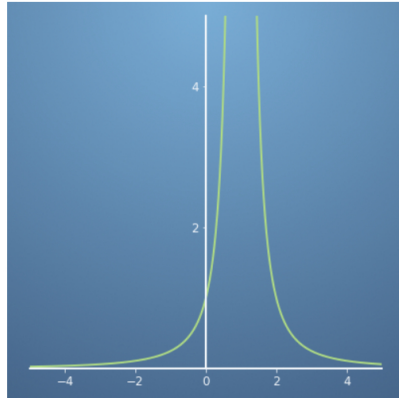


3. By finding the first three terms of the Taylor series shown above for the function $f(x) = \ln(x)$ (green line) about $x = 10$, determine the magnitude of the difference of using the second order Taylor expansion against the first order Taylor expansion when approximating to find the value of $f(2)$.

- ☒ $\Delta f(2) = 0.32$
- ☐ $\Delta f(2) = 1.0$
- ☐ $\Delta f(2) = 0$
- ☐ $\Delta f(2) = 0.5$

4. In some cases, a Taylor series can be expressed in a general equation that allows us to find a particular n^{th} term of our series. For example the function $f(x) = e^x$ has the general equation $f(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}$. Therefore if we want to find the 3^{rd} term in our Taylor series, substituting $n = 2$ into the general equation gives us the term $\frac{x^2}{2}$. We know the Taylor series of the function e^x is $f(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \dots$. Now let us try a further working example of using general equations with Taylor series.

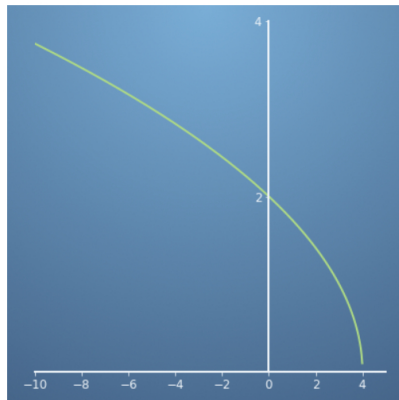
1 point



By evaluating the function $f(x) = \frac{1}{(1-x)^2}$ about the origin $x = 0$, determine which general equation for the n^{th} order term correctly represents $f(x)$.

- ☒ $f(x) = \sum_{n=0}^{\infty} (1+n)x^n$
- ☐ $f(x) = \sum_{n=0}^{\infty} (1+2n)(x)^n$
- ☐ $f(x) = \sum_{n=0}^{\infty} (2+n)(x)^n$
- ☐ $f(x) = \sum_{n=0}^{\infty} (1+n)(-x)^n$

1 point



5. By evaluating the function $f(x) = \sqrt{4-x}$ at $x = 0$, find the quadratic equation that approximates this function.

- ☒ $f(x) = 2 - \frac{x}{4} - \frac{x^2}{64} \dots$
- ☐ $f(x) = 2 + x + x^2 \dots$
- ☐ $f(x) = \frac{x}{4} - \frac{x^2}{64} \dots$
- ☐ $f(x) = 2 - x - \frac{x^3}{64} \dots$